Robert Kimball

kimball.r@northeastern.edu | 781.540.4491 | linkedin.com/in/robert-kimball-252b822a9 | robertlkimball.com
Boston, MA | Open to relocation | Available May 2027

Education

Northeastern University, Boston, MA *May 2027*
B.S. in Mechanical Engineering GPA: 3.88/4.0 | Dean's List | Tau Beta Pi | Pi Tau Sigma
Relevant Courses: Mechanical Engineering Design, Mechanics of Materials, Measurement & Analysis

Skills

CAD & Analysis: SolidWorks (CSWA), SolidWorks Simulation, Abaqus/MATLAB (coursework), hand calcs
Build & Test: 3D printing, Arduino C++/DAQ, thermocouples, strain gauges, pressure transducers, power tools
Manufacturing: DFM/DFMA, injection-molding defect and tooling review, root-cause analysis, supplier quotes

Experience

Hasbro, Pawtucket, RI
Project Engineer Co-op (Fashion, Preschool, Disney) *Jun 2026 – Dec 2026*

- Own engineering for 10 My Little Pony items (avg. ~100k units each): write the feasibility specs (target cost, tooling estimate, ex-factory dates) that launch each item with overseas teams, and present at all 3 stage gates
- Proposed an error-proofing DFMA change for a new collectible, adopted into the design: a thick ABS base frame for the common tier and stacked PVC layers for higher tiers, so every tier aligns the same way and assembles only one way
- Review ~50 injection-molded figure samples a month for function, molding defects (short shots, flash, sink) and screen accuracy vs. licensor specs; track issues with the overseas team to sign-off

Project Engineer Co-op (Games, Play-Doh, NERF) *Jun 2025 – Dec 2025*

- Root-caused cracking in PET vacuum-formed trays (stiff corners, edge-weakening patterns); new standard (larger radii, thinner gauge, corrugated corner protectors) passed QA drop tests, est. ~$120k/yr on one title
- Led a 5-person co-op team from concept to a well-received 3D-printed toy prototype; designed its spring-loaded rack-and-pinion slide and packaged electronics from torn-down products into PLA parts
- Self-initiated a cost-estimating tool (Power Apps, Power Query, SharePoint) built on a database of past quotes; cut a typical estimate from ~1 hour to ~5 minutes for the 4 games engineers

Project Engineer Co-op | ***Beacon Consulting Group***, Medford, MA *Jul 2024 – Dec 2024*

- Analyzed cost-to-complete on commercial construction projects; wrote VBA macros used across the company

Junior Designer & CAD Operator | ***Davey Engineering & Construction***, Canton, MA *Jun 2021 – Dec 2023*

- Modeled sprinkler, plumbing and HVAC systems (Bentley OpenBuildings, 2D/3D) for ~30 residential projects
- Ran NFPA 13/13D/13R sprinkler hydraulic calcs in Elite Software FIRE, checking demand against water supply; prepared ~30 permit sets and fire protection narratives citing NFPA and Massachusetts code

Projects

Animatronic Eye Mechanism | *SolidWorks, 3D Printing, Arduino C++* *Feb – Mar 2026*

- Designed a 3D-printed animatronic eye: a 12-tooth, module-0.8 gear pair couples both lids so one SG90 servo blinks them and a second sweeps the eye 100°; tuned gears and pivot clearance over 3 versions on a printed test stand

BenchPress Hand-Crank Folding Picnic Table – ME Design (5-person team) *Summer 2026*

- Hand-calculated stress and modified-Goodman fatigue (0–200 lb sit/stand cycles) for the steel frame, which selected AISI 1006 HR steel; sized the frame-to-brace bolt in shear (Grade 1, SF 4.8); modeled the parts in SolidWorks

Robotic Treasure Chest | *SolidWorks, 3D Printing, Arduino* *Dec 2025 – Jan 2026*

- Designed a servo-driven 3D-printed chest (70 × 75 × 67 mm) whose lid opens through a 12-tooth 1:1 bevel-gear pair at 90°; enlarged the gears (Ø12 → Ø16.1 mm, 45° cones) and replaced the bolt hinge with integral pins and hooks

Home Builds | *Power tools, framing, plumbing*

- Built a shower solo to Massachusetts plumbing code (framing, PVC liner, mortar bed pitched with a self-built 2×4 screed jig, PEX supply, tile); also framed and wired a kitchen and built a staircase

Measurement & Analysis Lab (ME 4505/4506) | *Arduino DAQ, Excel* *Spring 2026*

- Calibrated sensors against reference standards (NIST weights, reference thermometer, laser meter) and computed accuracy, precision and uncertainty; measured thermocouple and RTD time constants (0.36, 4.9 s)
- Characterized strain gauges in quarter-, half- and full-bridge circuits: hysteresis, drift, 30-trial repeatability with chi-squared normality and ANOVA, and the sample size needed for 1% precision (N = 59)